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# **PROJECT REPORT ON Hospital Management System**

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## Introduction

A **Hospital Management System (HMS)** is a computerized solution designed to streamline and manage the daily operations of a hospital. It focuses on automating the key activities of a hospital, including patient registration, appointment scheduling, billing, and record management.

By using a **Database Management System (DBMS)**, the project ensures data is stored efficiently, securely, and is easily accessible when needed. It replaces traditional paper-based systems with digital records, reducing manual errors, saving time, and improving service delivery.

### **Who will be users?**

The system will have Different types of users:

1. Patient: The Patient is the central entity in the hospital system. They are the individuals receiving medical care and treatment.
2. Doctor: A Doctor is a medical professional who provides care and treatment to patients.
3. Employee (including Receptionist and others): Employees in the hospital include Receptionists and other administrative staff.
4. Receptionists – Administrative staff managing patient appointments and records.

5. Management System – The central entity connecting and coordinating all users and their interactions within the hospital system.

**Future Scope:**

- Artificial Intelligence (AI) can be integrated into the Doctor entity for diagnosis and treatment recommendation
- Develop mobile applications for both patients and employees for easier access to records, appointments, and notifications.
- Integrate Internet of Things (IoT) devices that allow for real-time patient monitoring.

**Objective:**

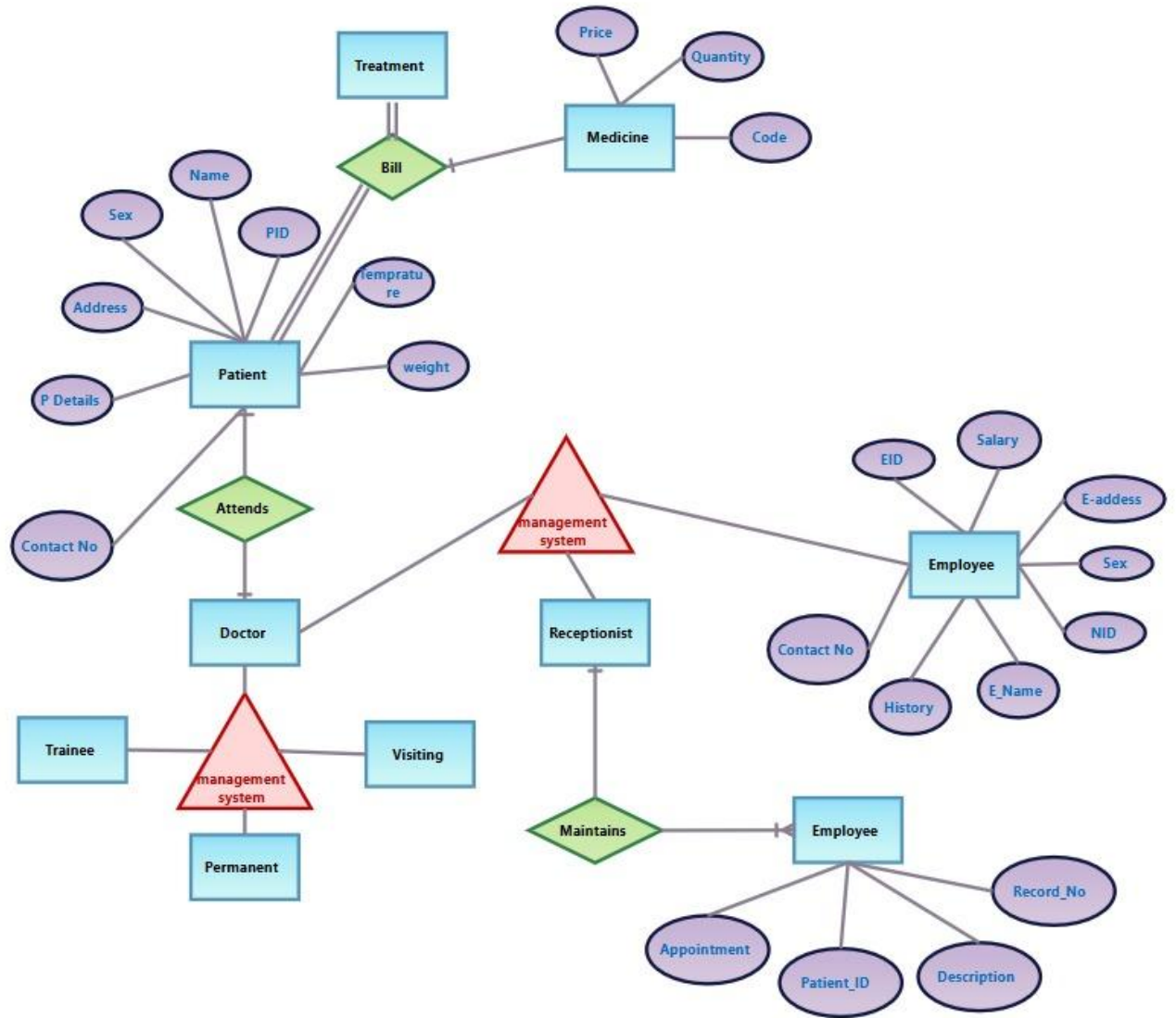
The objective of the ER diagram for the Hospital Management System (HMS) shown in the image is to visually represent the structure and relationships between various entities involved in the hospital's operations. The key objectives of the system can be broken down as follows:

**Key Relationships:**

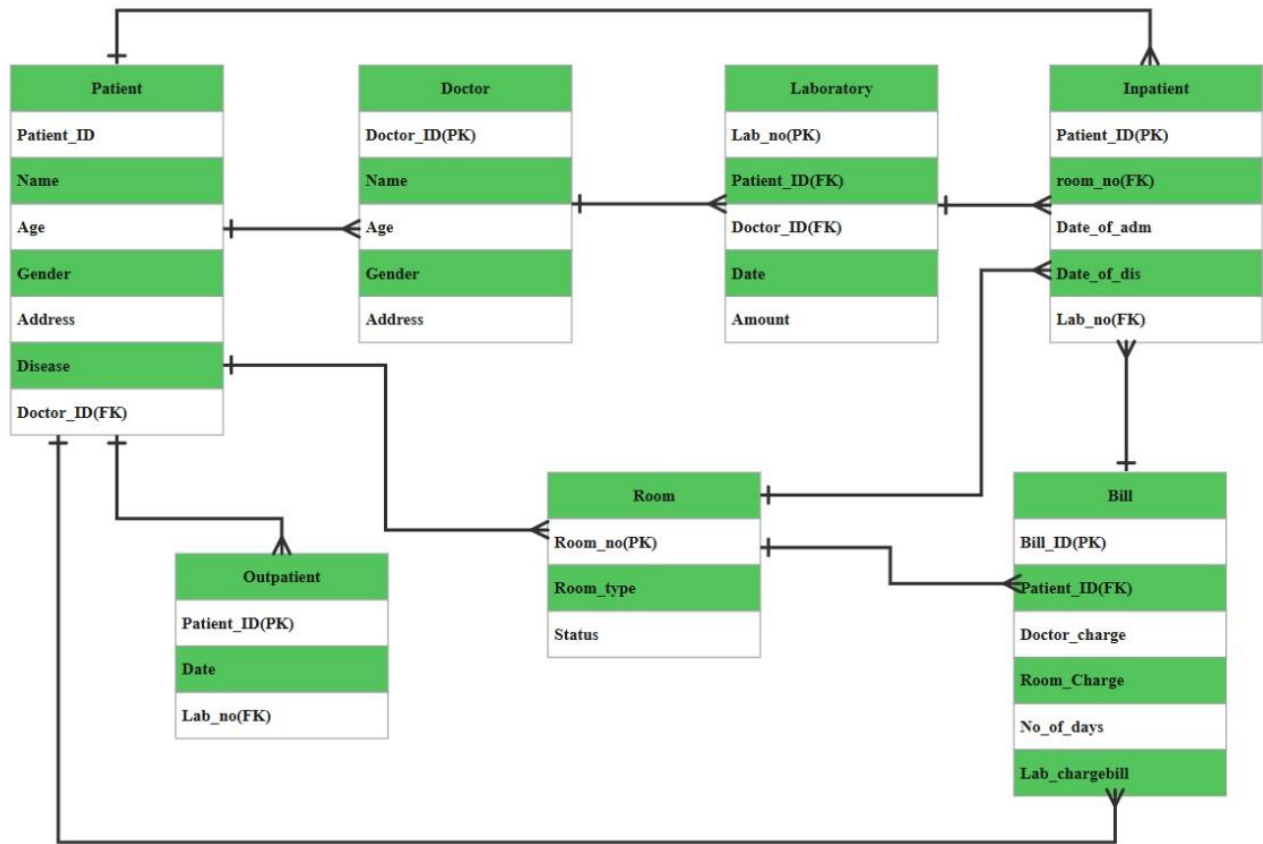
1. Attends: The Doctor attends to the Patient for treatments, consultations, or check-ups.
2. Maintains: The Receptionist maintains records related to the Employees and Patient appointments.
3. Management System: The Management System connects all entities, ensuring smooth operations and data flow across the hospital.

# ER DIAGRAM

ER Diagram for Hospital Management System



# RELATIONAL SCHEMA



# SQL Queries

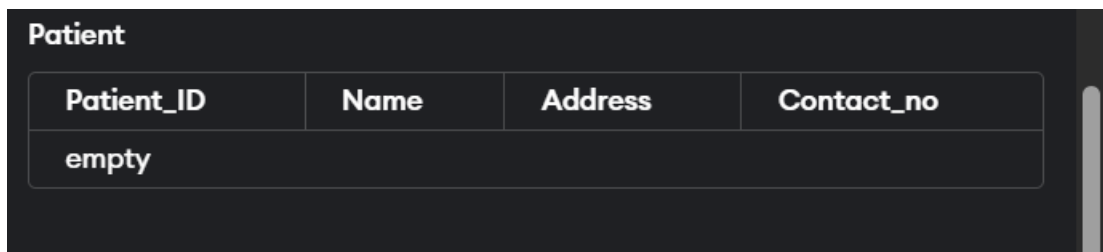
## 1. DDL(Data Definition Language) Queries

These queries create the database schema.

### Create Table:

```
CREATE TABLE Patient (  
    Patient_ID INT PRIMARY KEY,  
    Name varchar(100),  
    Address varchar(10),  
    Contact_no INT  
);
```

### Output:



The screenshot shows a database table named 'Patient'. The table has four columns: Patient\_ID, Name, Address, and Contact\_no. Below the column headers, there is a single row with the word 'empty' in the first column, indicating that the table is currently empty.

| Patient_ID | Name | Address | Contact_no |
|------------|------|---------|------------|
| empty      |      |         |            |

```
CREATE TABLE Doctor (  
    Doctor_ID INT PRIMARY KEY,  
    Name varchar(255),  
    Specialty varchar(100),  
    Contact_No INT  
);
```

## Output:

| Doctor    |      |           |            |
|-----------|------|-----------|------------|
| Doctor_ID | Name | Specialty | Contact_No |
| empty     |      |           |            |

```
CREATE TABLE Employee (  
    Employee_Id INT PRIMARY KEY,  
    E_Name varchar(255),  
    Sex varchar(100),  
    E_Address varchar(255),  
    Contact_no INT  
);
```

## Output:

| Employee    |        |     |           |            |
|-------------|--------|-----|-----------|------------|
| Employee_Id | E_Name | Sex | E_Address | Contact_no |
| empty       |        |     |           |            |

## 2.DML(Data Manipulation Language) Queries:

These queries insert, delete and update data.

## Insert Data

INSERT INTO

Patient(Patient\_ID,Name,Address,Contact\_no)

VALUES(101,'John','Mumbai',9933203452);

VALUES(102,'Joe','Delhi',9937260123);

VALUES(103,'Smith','Delhi',995410034);

## Output:

| Patient    |       |         |            |
|------------|-------|---------|------------|
| Patient_ID | Name  | Address | Contact_no |
| 101        | John  | Mumbai  | 9933203452 |
| 102        | Joe   | Delhi   | 9937260123 |
| 103        | Smith | Delhi   | 995410034  |

INSERT INTO

Doctor(Doctor\_ID,Name,Specialty,Contact\_No)

VALUES(201,'Robert','Cardiologist',9832012340);

VALUES(202,'Betty','Dentist',9882255743);

VALUES(203,'Robinson','Orthopadic',9772340123);



## Output:

| Doctor    |          |              |            |
|-----------|----------|--------------|------------|
| Doctor_ID | Name     | Specialty    | Contact_No |
| 201       | Robert   | Cardiologist | 9832012340 |
| 202       | Betty    | Dentist      | 9882255743 |
| 203       | Robinson | Orthopadic   | 9772340123 |

```
INSERT INTO
Employee(Employee_Id,E_Name,Sex,E_Address,
Contact_no)
VALUES(1,'Sahil','Male','Delhi',9988824501);
VALUES(2,'Priya','Female','Mumbai',9912035432);
VALUES(3,'Sakshi','Female','Patna',9285058745);
VALUES(4,'Rohit','Male','Gujarat',9988753021);
```

## Output:

| Employee    |        |        |           |            |
|-------------|--------|--------|-----------|------------|
| Employee_Id | E_Name | Sex    | E_Address | Contact_no |
| 1           | Sahil  | Male   | Delhi     | 9988824501 |
| 2           | Priya  | Female | Mumbai    | 9912035432 |
| 3           | Sakshi | Female | Patna     | 9285058745 |
| 4           | Rohit  | Male   | Gujarat   | 9988753021 |

## Update Data

Modify existing records:

```
UPDATE Employee
```

```
Set E_name = 'Mohit'
```

```
WHERE Employee_Id = 1;
```

**Output:**

| Employee    |        |        |           |            |
|-------------|--------|--------|-----------|------------|
| Employee_Id | E_Name | Sex    | E_Address | Contact_no |
| 1           | Mohit  | Male   | Delhi     | 9988824501 |
| 2           | Priya  | Female | Mumbai    | 9912035432 |
| 3           | Sakshi | Female | Patna     | 9285058745 |
| 4           | Rohit  | Male   | Gujarat   | 9988753021 |

## Delete Data

Remove Specific Records.

```
DELETE From Employee
```

```
where Employee_ID = 4;
```

**Output:**

| Employee    |        |        |           |            |
|-------------|--------|--------|-----------|------------|
| Employee_Id | E_Name | Sex    | E_Address | Contact_no |
| 1           | Mohit  | Male   | Delhi     | 9988824501 |
| 2           | Priya  | Female | Mumbai    | 9912035432 |
| 3           | Sakshi | Female | Patna     | 9285058745 |

## **Conclusion for the Hospital Management System**

The Entity-Relationship (ER) diagram for the Hospital Management System provides a clear and organized representation of the system's data structure. It effectively outlines the relationships between key entities such as Patients, Doctors, Appointments, Staff, Wards, and Treatments. This model ensures efficient data handling, reduces redundancy, and improves data integrity. By mapping out these entities and their interconnections, the ER diagram serves as a foundational blueprint for designing a robust and scalable hospital management database system. It supports better decision-making, streamlines hospital operations, and ultimately enhances the quality of healthcare services delivered.

